Simplex-Based Open-Set Recognition Beyond the Simplex Regime: Theory, Threshold Analysis, and Low-Dimensional Extensions of UCDSC

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YouTube: `[link placeholder]` **GitHub:** `https://github.com/mayank172900/ucdsc-phase2-project`

1 Base Paper Overview

1.1 Open-Set Recognition in Medical Imaging

Open-set recognition (OSR) requires a classifier to simultaneously identify samples from known training classes and reject inputs from previously unseen categories at test time [1]. In medical imaging, this capability is safety-critical: a model that confidently assigns a known-class label to a truly novel pathology risks misdiagnosis with potentially severe clinical consequences. Unlike closed-set classification, OSR explicitly models the open-world nature of real deployments through novelty detection mechanisms.

1.2 The UCDSC Framework

The Uncertainty-aware Deep Simplex Classifier (UCDSC) [2] builds on the Deep Simplex Classifier (DSC) [3], which fixes class prototypes at the vertices of a simplex equiangular tight frame (ETF), motivated by the neural collapse phenomenon [4]. For an embedding $z \in \mathbb{R}^d$ with C known classes having prototypes $\{s_1, \ldots, s_C\}$, the uncertainty score is defined as

$$U(z) = \frac{d_{\min}(z)}{\mu(z)}, \quad d_{\min}(z) = \min_j \|z - s_j\|, \qquad (1)$$

where $\mu(z)$ averages the distances to all non-nearest prototypes. A sample is rejected when $U(z) > \theta$ for a threshold $\theta \in [0, 1]$.

UCDSC trains with a composite loss $L_{\text{total}} = L_{\text{intra}} + \lambda_o L_o + \lambda_u L_u$, where L_{intra} encourages feature-prototype clustering, L_o is a triplet-based outlier-exposure term, and L_u regularizes mean uncertainty on training features. The method achieves strong performance across multiple MedMNIST benchmarks [5] and the Augmented Skin Conditions dataset.

1.3 Motivation: The Low-Dimensional Regime

The simplex ETF construction requires $d \geq C - 1$ to embed a regular simplex in $\mathbb{R}^d$. While this is typically satisfied in standard training (e.g., $d = 512$), practical constraints—model compression, privacy-preserving low-dimensional embeddings, or high class counts relative to embedding size—can push the system into the regime $d < C - 1$. In this regime, perfect simplex symmetry cannot be realized.

1.4 Theoretical Lens

Our companion theoretical paper [6] develops a unified framework for simplex-ratio OSR that operates in all embedding dimensions, including $d < C - 1$. The key theoretical contributions relevant to this report are:

- **General Lipschitz analysis:** For arbitrary distinct prototypes, U is globally Lipschitz with constant $4C/((C-1)\Delta)$ where Δ is the minimum inter-prototype distance, and U-acceptance regions are compact unions of prototype-centered balls with explicit radii (Theorem 2 of [6]).
- **Balanced equal-norm codes:** For codes satisfying $\|s_j\| = R$ and $\sum_j s_j = 0$—which exist in every $d \geq 2$—the squared score U_2 has sublevel sets that are exact unions of Euclidean balls (Theorem 8 of [6]).
- **Sharp dichotomy:** A simplex-defect parameter $\lambda_j = \sqrt{B_j}/A_j \geq 1$ equals unity for all j if and only if the prototypes form a regular simplex, which requires $d \geq C-1$ (Theorem 10 of [6]). Below this threshold, $\lambda_j > 1$ quantifies *graceful degradation*, not collapse.
- **FAR bounds:** Under mild distributional assumptions, the false acceptance rate decays exponentially in d, with explicit dependence on the defect parameter (Theorem 12 of [6]).

The sharp dichotomy theorem is central to our experimental program: it predicts that reducing d below $C - 1$ should degrade—but not destroy—rejection performance,

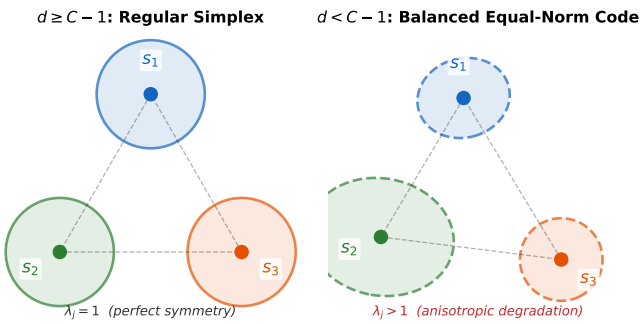

Figure 1: Rejection geometry in the simplex ($d \geq C-1$) vs. sub-simplex ($d < C-1$) regime. Left: regular simplex prototypes yield symmetric, equal-radius acceptance balls ($\lambda_j = 1$). Right: when $d < C-1$, a balanced equal-norm code replaces the simplex; the defect $\lambda_j > 1$ induces anisotropic, non-uniform acceptance regions.

and that the severity of degradation is governed by λ_j. Figure 1 illustrates this contrast.

2 Our Approach and Contribution

Our project combines paper-aligned reproduction, threshold analysis, and theory-motivated extension experiments. We describe each component below.

2.1 Reproduced Baselines

We reproduced the UCDSC training and evaluation pipeline on two MedMNIST datasets used in the original paper: **BloodMNIST** (8 known classes, cell-type classification from blood smear images) and **DermaMNIST** (7 known classes, dermatological lesion classification). These reproductions serve as our reference baselines for closed-set accuracy (ACC), area under the ROC curve (AUROC), open-set classification rate (OSCR), true negative rate (TNR), and decision-threshold accuracy (DTACC). We report means and standard deviations across multiple data splits to account for variance.

2.2 Threshold Calibration and Operating-Point Analysis

The original UCDSC paper reports performance at fixed operating points, but does not systematically study the tradeoff between known-class recall and false acceptance rate. We conduct a threshold calibration study using four complementary metrics: false acceptance rate (FAR), known-class rejection rate (KRR), true negative rate (TNR), and decision-threshold accuracy (DTACC). Specifically, we compare two calibration strategies: (i) a recall-controlled threshold targeting ∼95% known-class recall, and (ii) the best-DTACC threshold that maximizes the combined known/unknown classification accuracy. This analysis directly connects to Theorem 12(i) of our theoretical framework [6], which provides population-level recall control via class-conditional quantile bounds under Assumption A5.

2.3 Low-Dimensional BloodMNIST

To empirically test the dimension-sensitivity predictions of the theory, we re-ran the BloodMNIST pipeline with the embedding dimension reduced from the default $d = 512$ to $d = 32$. While $d = 32$ still satisfies $d \geq C - 1 = 7$, the sharp reduction probes whether the FAR bounds that scale exponentially in d (Theorem 12 of [6]) manifest as observable degradation in practice.

2.4 CIFAR-100 Extension

To test generalizability beyond the medical domain, we ran UCDSC on CIFAR-100 with a known/unknown class split. This extension is not part of the original UCDSC evaluation and tests whether the simplex-based rejection geometry transfers to natural image distributions with higher class counts and greater visual diversity.

2.5 PathMNIST: Cancer-Domain $d < C - 1$ Extension

Our most theory-motivated experiment applies UCDSC to PathMNIST (colorectal cancer histopathology) with $C = 6$ known classes and embedding dimension $d = 4$, placing the system firmly in the $d < C - 1$ regime since $4 < 5$. By the sharp dichotomy theorem (Theorem 10 of [6]), the prototypes cannot form a regular simplex and the defect parameters satisfy $\max_j \lambda_j > 1$, predicting degraded but non-trivial rejection geometry. This experiment required debugging and correcting several implementation issues in the UCDSC pipeline to support non-standard embedding dimensions end-to-end, including modifications to the prototype initialization, feature extractor output layer, and evaluation routines.

We emphasize that PathMNIST is an *exploratory extension experiment*, not a reproduction of any result from the original paper.

3 Experiment Results

3.1 Reproduced Baselines

Table 1 summarizes the reproduced UCDSC baselines on BloodMNIST and DermaMNIST. BloodMNIST exhibits strong closed-set accuracy (ACC ≈ 98.2%) and moderate open-set discrimination (AUROC ≈ 81.1%). DermaMNIST, a harder dataset with more visual overlap between classes, shows lower accuracy and substantially weaker open-set performance, particularly in TNR (14.5% vs. 34.6%). These results are broadly consistent with the difficulty hierarchy reported in the original UCDSC paper and confirm that the reproduced pipeline is functioning correctly.

3.2 Threshold Calibration

Table 2 presents the FAR at two operating points: a recall-controlled threshold targeting ∼95% known-class recall, and the best-DTACC threshold.

Table 1: Reproduced UCDSC baselines (mean ± std).

Metric	BloodMNIST	DermaMNIST
ACC (%)	98.242 ± 0.662	84.414 ± 6.389
AUROC (%)	81.122 ± 8.230	67.373 ± 7.903
OSCR (%)	80.544 ± 8.193	60.133 ± 11.355
TNR (%)	34.601 ± 18.037	14.545 ± 5.641
DTACC (%)	76.820 ± 5.796	68.146 ± 3.255

Table 2: FAR at two operating points (mean ± std).

Operating Point	BloodMNIST	DermaMNIST
Recall ∼95% (FAR)	0.652 ± 0.180	0.855 ± 0.056
Best-DTACC (FAR)	0.082 ± 0.026	0.244 ± 0.225

The results reveal a pronounced tradeoff: maintaining high known-class recall forces the threshold into a regime where the majority of unknown samples are incorrectly accepted (FAR > 0.65 on BloodMNIST and > 0.85 on DermaMNIST). Relaxing the recall target to the best-DTACC operating point reduces FAR dramatically—by roughly 8× on BloodMNIST—while trading off some recall. DermaMNIST is substantially harsher in this tradeoff, with best-DTACC FAR still at 0.244 and high variance, reflecting the intrinsic difficulty of separating known from unknown in this skin-lesion setting.

This analysis underscores that threshold selection is a genuine operating decision with material impact on clinical utility, not a cosmetic post-processing step (see Figure 2, left panel). The theoretical framework provides tools for principled threshold selection via the quantile-control corollary (Theorem 12(i) of [6]) and the practical Algorithm 1 for estimating the rescaled score-tail parameter σ from validation data.

3.3 Low-Dimensional BloodMNIST

Table 3 compares the default ($d = 512$) and reduced ($d = 32$) embedding dimensions on BloodMNIST.

Reducing the embedding dimension from 512 to 32 causes moderate degradation in AUROC (−3.0 points) and OSCR (−2.7 points), but a much more substantial worsening of the best-DTACC FAR (0.082 → 0.289, a ∼3.5× increase). The increased variance in the $d = 32$ setting (±12.8 vs. ±8.2 for AUROC) further signals instability at lower dimensions. These findings are consistent with the theoretical prediction that FAR bounds degrade as embedding dimension decreases: the exponential scaling FAR $\lesssim C\exp(-c(\theta)d)$ from Theorem 12 of [6] implies that halving d substantially loosens the bound. While $d = 32$ still exceeds $C - 1 = 7$ (so perfect simplex symmetry is available), the practical effect on the rejection operating point is already pronounced.

3.4 CIFAR-100 Extension

Table 4 reports UCDSC performance on CIFAR-100 under the default and normalized/b9 configurations.

CIFAR-100 results are notably strong: ACC exceeds 94%, AUROC exceeds 92%, and TNR reaches ∼60%—all substantially above the medical baselines. The normalized variant yields marginal improvements across all metrics. These results suggest that the simplex-based rejection geometry of UCDSC generalizes effectively beyond the medical imaging domain for which it was originally designed, supporting broader applicability of the framework. Figure 3 contrasts these results with the PathMNIST extension.

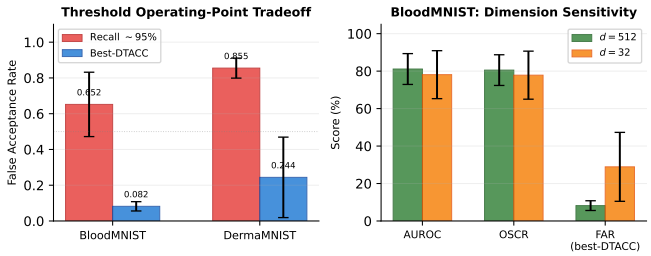

Figure 2: Left: FAR at two operating points—the recall-∼95% threshold admits most unknowns, while the best-DTACC threshold dramatically reduces FAR. Right: reducing embedding dimension from 512 to 32 on BloodMNIST degrades AUROC/OSCR moderately but worsens the best-DTACC FAR by ∼3.5×.

Table 3: BloodMNIST: default vs. reduced dimension.

Metric	$d = 512$	$d = 32$
AUROC (%)	81.122 ± 8.230	78.108 ± 12.801
OSCR (%)	80.544 ± 8.193	77.839 ± 12.824
Best-DTACC FAR	0.082 ± 0.026	0.289 ± 0.184

3.5 PathMNIST: $d < C - 1$ Extension

Table 5 presents the PathMNIST results with $C = 6$ known classes and $d = 4$ ($d < C - 1 = 5$).

Performance is weak and highly unstable across all metrics. ACC of ∼43% is far below the ceiling for this dataset, AUROC falls below 40%, and TNR is near-zero (∼2%), indicating that the model almost never correctly rejects unknown samples. The high standard deviations (e.g., ±13.6 for AUROC) reflect substantial run-to-run variability, consistent with the prediction that low-dimensional prototype configurations induce anisotropic, unstable rejection regions. The normalized variant does not remedy the situation and in fact worsens AUROC.

These results are qualitatively consistent with the sharp dichotomy theorem: at $d = 4 < C-1 = 5$, perfect simplex symmetry is unavailable, $\max_j \lambda_j > 1$, and the resulting prototype configuration induces a defective rejection geometry. The theory predicts degradation governed by the defect parameter λ_j, and the empirical outcome—severe but not complete collapse—aligns with this prediction. We note, however, that the PathMNIST experiment confounds the dimensional effect with the intrinsic difficulty of the cancer-domain classification task at such low capacity, and we therefore present this as an exploratory finding rather than a clean isolation of the theoretical mechanism.

Table 4: CIFAR-100 extension results (mean ± std).

Metric	Default	Normalized / b9
ACC (%)	94.53 ± 1.00	94.70 ± 1.02
AUROC (%)	92.03 ± 2.06	92.07 ± 2.08
OSCR (%)	87.95 ± 1.71	88.11 ± 1.64
TNR (%)	59.33 ± 7.81	60.34 ± 7.22

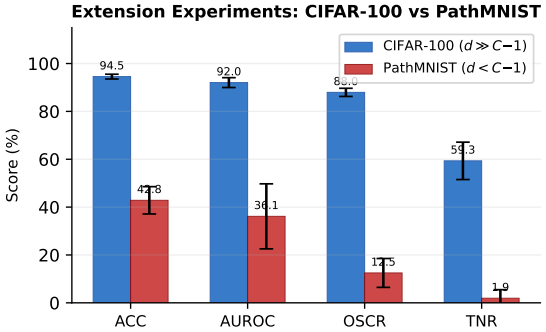

Figure 3: Extension experiments: CIFAR-100 ($d \gg C-1$) vs. PathMNIST ($d = 4 < C-1 = 5$). The stark performance gap is consistent with the sharp dichotomy prediction that rejection geometry degrades when $d < C-1$.

Conclusion

Our reproduced baselines on BloodMNIST and DermaMNIST confirm the original UCDSC claims within expected variance. The threshold calibration analysis shows that operating-point selection is a practically consequential decision, linked to the quantile-control and FAR-bound machinery in [6]. The low-dimensional BloodMNIST experiment provides empirical evidence for dimension sensitivity, consistent with exponential-in-d FAR scaling (Theorem 12). CIFAR-100 suggests the simplex rejection geometry generalizes beyond medical imaging to natural image domains.

The PathMNIST extension provides a theory-aligned exploratory data point in the $d < C-1$ regime. The sharp dichotomy theorem (Theorem 10) predicts that below the simplex threshold, rejection geometry survives with degradation governed by λ_j, and the observed weak but non-zero PathMNIST performance is consistent with this. More controlled experiments isolating the dimensional effect from task difficulty are needed, but the present results support practical relevance of the framework across both threshold and dimension axes.

Taken together, the theoretical framework in [6] and the experiments here provide a more complete picture of simplex-based open-set recognition: strong performance within the intended regime, principled threshold calibration tools, and a clear theoretical account of what degrades when the classical simplex assumption is relaxed.

Table 5: PathMNIST (d=4, C=6) extension (mean ± std).

Metric	Default	Normalized / b9
ACC (%)	42.840 ± 5.705	42.840 ± 5.705
AUROC (%)	36.141 ± 13.591	30.964 ± 16.765
OSCR (%)	12.491 ± 6.034	13.593 ± 6.234
TNR (%)	1.923 ± 3.582	1.902 ± 3.537

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